

One-Page Pitch: Belief-Maintenance-Aware LoRa Reporting

A trace-replay lab study connecting low-power IoT delivery metrics to information usefulness

Core idea: Low-power sensing networks are usually judged by packet delivery, latency, and energy use. This project asks a second question: did the packets that arrived actually help maintain the downstream belief state? A synthetic belief-demand trace generator supplies candidate observations and usefulness metadata; approximately 30 ESP32 LoRa boards provide the physical communication layer.

Proposed proof-of-concept experiment

Question	When LoRa communication opportunity is scarce, which reporting policy preserves belief quality best per transmitted or delivered packet?
Method	Generate synthetic candidate observations for about 30 logical sensing nodes; assign each packet a priority, expected-usefulness score, and stale-after threshold; replay selected packets over ESP32 LoRa boards.
Scarcity model	Do not rely on naturally overloading LoRa. Instead impose a fixed reporting budget per time window, modelling scarce airtime, energy, duty-cycle, or reporting opportunity.
Policies	Compare periodic reporting, event-triggered reporting, random budgeted reporting, uncertainty-prioritized reporting, and a simple belief-maintenance-aware policy.
Physical data	Receiver/gateway logs packet identifiers, receive times, missing sequence numbers, RSSI, SNR, and eventually energy or airtime estimates.
Analysis	Join receiver logs with trace metadata to compare ordinary communication metrics against usefulness and belief-maintenance metrics.

Why this is scientifically interesting

- The result is not just another LoRa demo: the contribution is an evaluation frame that separates what arrived from what mattered.
- A publishable separation would show that policies with similar packet delivery ratio can produce different useful-update rates or entropy reduction per delivered packet.
- The physical testbed supplies timing irregularity, loss, contention, receiver effects, and realistic low-power constraints that a simulator should not claim by itself.
- The scope is intentionally bounded: this is a repeatable lab method for environmental IoT sensing, not an operational wildfire deployment.

Primary metrics

Communication	Usefulness	Belief / cost
Packet delivery ratio; latency; inter-arrival gaps; missing sequence fraction; RSSI/SNR	Useful-update rate; stale packet fraction; missed high-priority packet fraction	Entropy reduction per delivered packet; belief improvement per packet; entropy reduction per joule if energy is measured

Immediate ask: Validate the LoRa-side assumptions: practical airtime budget, radio settings, logging design, and whether the first study should emphasize constrained airtime, duty cycle, energy, contention, or a combination.

Draft concept for DCOSS-IoT / low-power IoT discussion